

# Youth Drug Use Prediction using Decision Trees

A Machine Learning Project Using the NSDUH 2023 Dataset

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## Summary

This project used survey data from the National Survey on Drug Use and Health (NSDUH, 2023) to explore the behavioral, demographic, and psychosocial factors influencing drug use among youths (under 18 years). The data included over 30,000 responses but I used only Youth Focused Data only which include around 10561 observations. This project sought to investigate which factors including behavioral, demographic, and psychosocial are associated with marijuana using predictive modelling. The purpose of this project, check drug use with Marijuana that was used to solve the below 3 machine learning problems.

The below work focuses on three key machine learning problem types, each tied to a specific question of interest. Our main goal was to understand how behavioral, peer, school, and family-related variables predict:

1. Whether a youth has ever used marijuana (binary classification)
2. How frequently youth use marijuana monthly (multi-class classification)
3. How many days per year youth report using marijuana (regression)

Through this predictive modelling the study observed insights about risk factors such as peer influence, parental attitudes, and school behaviors on marijuana use.

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## 1. Project Introduction

### 1.1 Background & Motivation

Substance use during adolescence is a major public health concern [through National Library of Medicine](#), associated with negative outcomes ranging from mental health disorders to educational failure. Policymakers and intervention programs benefit from identifying the socio-environmental factors that most influence youth behavior.

This project used decision trees and tree-based ensemble models, chosen for their interpretability and strong performance on mixed-type survey data. All models are built using R.

## 1.2 Objectives

- Explore marijuana use behavior using supervised learning models on NSDUH youth data.
  - Apply decision trees, bagging, random forests, and boosting where appropriate.
  - Identify key predictive features and provide interpretable outputs.
  - Evaluate models on performance metrics (accuracy, MSE).
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## 2. Data Overview

### 2.1 Dataset Description

I use a preprocessed subset of the 2023 NSDUH dataset focused on youth under 18 years of age. The original dataset includes over 30,000 respondents and nearly 3,000 variables. The youth subset ([youth data.xlsx](#)) includes 79 curated variables related to drug use, school behavior, peer influence, and parental guidance.

### 2.2 Selected Target Variables

Majorly focusing on specific target variable drug type (marijuana) in each problem.

| Problem Type               | Target Variable | Description                                                      |
|----------------------------|-----------------|------------------------------------------------------------------|
| Binary Classification      | MRJFLAG         | Has ever used marijuana (0 = No, 1 = Yes)                        |
| Multi-Class Classification | MRJMDAYS        | Monthly use frequency of marijuana (Seldom, Sometimes, Frequent) |
| Regression                 | IRMJFY          | Number of days marijuana used in the past year.                  |

### 2.3 Predictor Variables

The dataset included around 60 predictor variables and some of them are

- What do your friend think about marijuana?
- Do your parents disapprove of you drinking?
- How many days did you skip school?

And various other predictor variables for check influence in Youth drug use.

## 2.4 Data Cleaning & Feature Engineering

- Variables that directly implied the target were removed to prevent any multicollinearity issue on each model, so created each new subset for 3 problems.
  - Outliers and special codes (e.g., 991 = “never used”) were carefully filtered.
  - All categorical target variables were converted to factor class.
  - Missing values were dropped or replaced with zero based on context and variable type.
  - Re-coded usage ranges into interpretable categories.
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## 3. Modeling Approach

### 3.1 Tools & Libraries

- tree, randomForest, and bagging, boosting for modeling
- dplyr, tidyverse, readxl for data processing
- Cross-validation and hyperparameter tuning applied selectively

### 3.2 Evaluation Metrics

| Task Type      | Metric                     |
|----------------|----------------------------|
| Classification | Accuracy, Confusion Matrix |
| Regression     | Mean Squared Error (MSE)   |

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## 4. Results by Problem Type

### 4.1 Binary Classification: Marijuana Use (MRJFLAG)

**Objective:** What factors predict whether a youth has ever used marijuana?

| Used Marijuana | Count |
|----------------|-------|
| No (0)         | 8971  |
| Yes (1)        | 1590  |

From here got to know that 85% of youth had never used marijuana , this might be consider as imbalance data

**Model Used:** Decision Tree (with pruning)

**Target:** MRJFLAG (0 or 1)

**Training Size:** 7,392 observations

**Test Accuracy:** 87.79%

**Unpruned and Pruned train error:** 12.46%

**Unpruned and Pruned test error :**12.21%

**Confusion Matrix:**

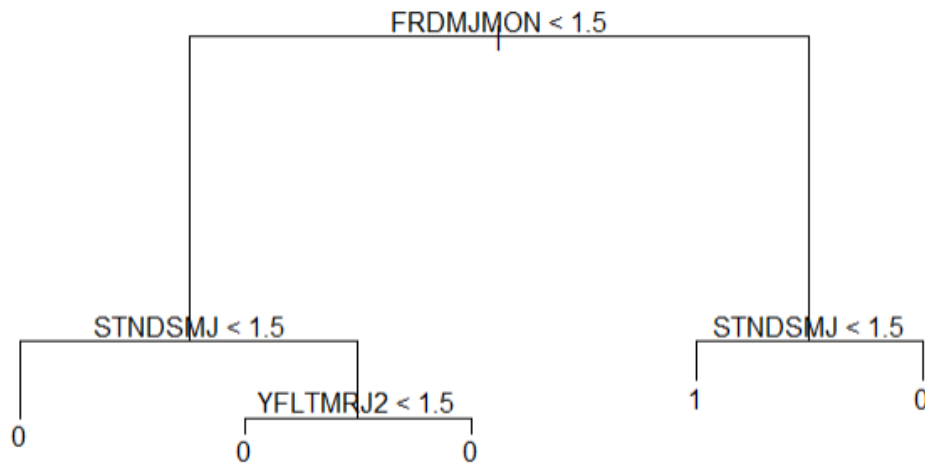
| Actual \ Predicted | 0    | 1   |
|--------------------|------|-----|
| 0                  | 2564 | 139 |
| 1                  | 248  | 218 |

**Key Predictors:**

- FRDMJMON: close friends feel about youth use marijuana mon.
- STNDSMJ: Perceived in certain grade usage at school of marijuana
- YFLTMRJ2: Youth's opinion on peer experimentation

**Tree:**

Youth who perceive their friends disapprove of marijuana are less likely to have ever used it. Peer norms play a dominant role in shaping behavior because above 3 factors has shown to have an impact on the use of marijuana.



#### Interpretation:

- Friends Disapprove Marijuana Use (FRDMJMON): Youth who believed their close friends disapproved had much lower likelihood of using marijuana.
- Peer Use Norms (STNDSMJ): If youth think most classmates use marijuana, they were more likely to use it themselves.
- Personal Attitudes (YFLTMRJ2): Youth who personally disapproved of trying marijuana were less likely to use.

## 4.2 Multi-Class Classification: Monthly Marijuana Use Frequency

**Objective and Target:** To classify youths into usage frequency categories for the past month (MRJMDAYS):

| Category  | Description     | Frequency |
|-----------|-----------------|-----------|
| Seldom    | 1–2 days/month  | 195       |
| Sometimes | 3–5 days/month  | 149       |
| Frequent  | 6–30 days/month | 408       |

**Model Used:** Decision Tree, Bagging, Random Forest, Boosting omitted due to performance and support warnings

**Training and Test Size:** 526 observations and 226 observations

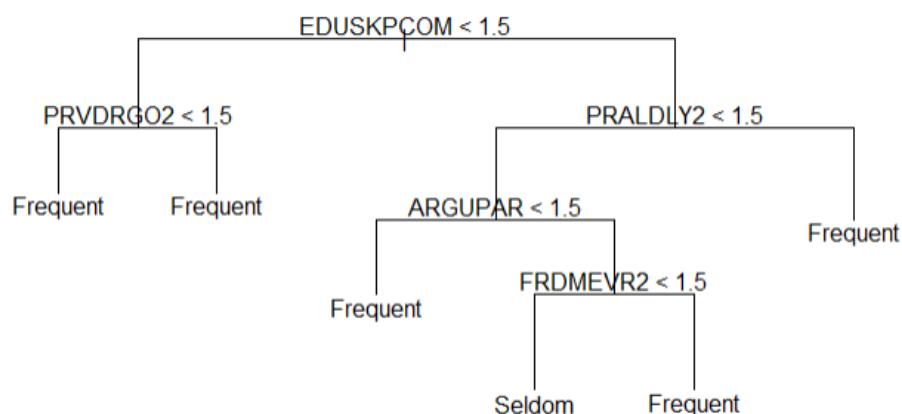
**Best Accuracy** (Random Forest): 48.23%

**Classification Tree test error:** 49.56 %

**Key Predictors:**

- EDUSKPCOM: Days skipped school
- PRALDLY2: Parental disapproval of daily alcohol use
- FRDMEVR2: Friends' opinions on trying marijuana
- PRVDRGO2: Participated in substance prev. program outside school
- ARGUPAR: Had a fight with parent in past year

**Tree:**



**Interpretation:**

after looking into tree structure and node .

- Skipped School (EDUSKPCOM): Youth missing more school tended to report more frequent use.
- Program Exposure (PRVDRGO2): Those not exposed to drug prevention programs were more likely to use frequently.
- Parental Disapproval (PRALDLY2): Protective effect observed again here, youth perceiving disapproval from parents were less likely to be frequent users.
- Also skipped school days argued with parents (ARGUPAR), or had peers who approved marijuana use (FRDMEVR2) were more likely to fall into the Frequent use category.

### 4.3 Regression: Annual Marijuana Use Days (IRMJFY)

**Objective:** To predict the total number of days marijuana was used in the past year (IRMJFY).

**Model Used:** Regression Tree, Random Forest along with tuned mtry

**Target:** IRMJFY (1 to 365)

**Training Size:** 934 observations

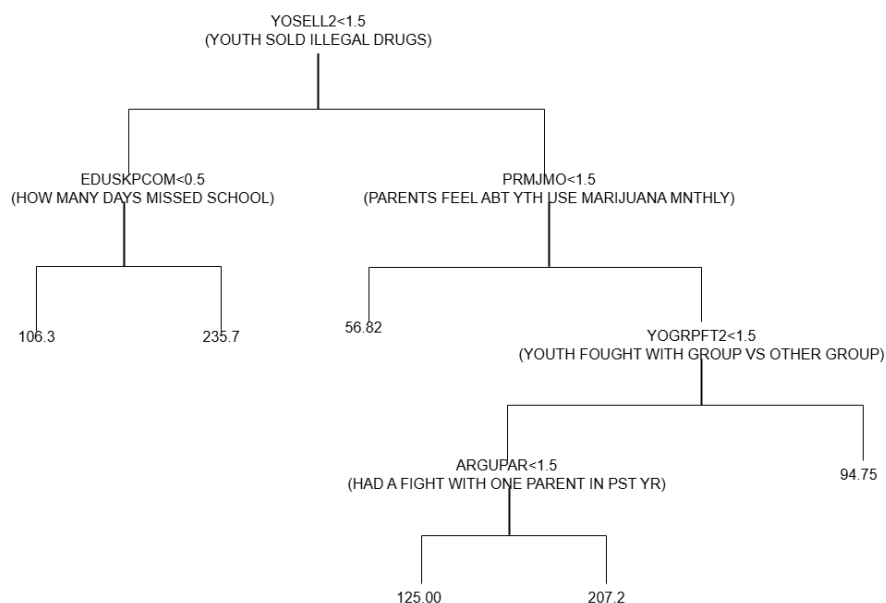
**Results:**

| Model             | Test MSE |
|-------------------|----------|
| Unpruned Tree     | 10280.03 |
| Pruned Tree       | 10157.17 |
| Random Forest     | 9671.08  |
| Tuned RF(mtry=10) | 9480.99  |

**Key Predictors:**

- YOSELL2: Whether youth has sold drugs
- EDUSKPCOM: Skipped school days
- PRMJMO: How youth perceive parental disapproval of monthly use
- YOGRPFT2: youth fought with group vs other group
- ARGUPAR: had a fight with parent in past year

**Tree :**



**Interpretation:**

Youth reporting drug sales or poor school attendance tend to have much higher marijuana use days annually. These behavioral indicators show strong associations with heavy use patterns. Which means Co-occurring risk behaviors like selling drugs and skipping school are strong predictors of annual marijuana use.

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## 5. Comparative Performance

### 5.1 Model Summary Table

| Model / Task     | Metric   | Score     |
|------------------|----------|-----------|
| Binary Tree      | Accuracy | 87.79%    |
| Multi-Class Tree | Accuracy | 50.44%    |
| Regression Tree  | MSE      | 10,618.77 |
| Pruned Tree      | MSE      | 10,157.17 |
| RF (Regression)  | MSE      | 9,222.91  |

### 5.2 Key Findings

- Peer influence is the most consistent predictor across all models.
  - Parental involvement showed a protective role against frequent or heavy drug use.
  - School attendance is a critical factor in predicting both frequency and intensity of substance use.
  - Missing school and fighting with parents are strong behavioral indicators of higher use frequency and intensity.
  - Multiclass classification presented more challenges than binary classification, partly due to class imbalance and overlapping behavioral features.
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## 6. Conclusion & Recommendations

Across all three tasks (binary, multi-class, regression), the models consistently identified a few core drivers of youth drug behavior of Close Friend Influence can be Strongest and most consistent predictor, Parental Involvements of disapproval can protect youth by using drug and School Involvement which skipping school has higher use of marijuana and this are behavioral risk indicators (e.g., skipping school, selling drugs)

### Recommendations:

- Prevention programs should focus on reinforcing peer-led anti-drug messaging.
  - Interventions can be targeted at youth who show early behavioral risk markers.
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## 7. References

- Trends in marijuana and heavy alcohol use by cigarette smoking status among US adults: An analysis of the 2002-2019 NSDUH: <https://www.sciencedirect.com/science/article/abs/pii/S0376871624002771>
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- Tidyverse & Tree Modeling Docs: <https://dplyr.tidyverse.org>, <https://www.rdocumentation.org>